Oxford College of Emory University

# Chapter 13: Fiscal Policy

## Part 1: Fiscal Policy and Stabilization

ECON_OX 112 · Principles of Macroeconomics

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# The Federal Budget: FY2025

**Federal government only · October 2024–September 2025 · Billions of dollars**

| Revenues | $ billions |
|---|---|
| Individual income taxes | 2,656 |
| Payroll taxes | 1,748 |
| Corporate income taxes | 452 |
| Customs duties (tariffs) | 195 |
| Excise taxes | 106 |
| Estate and gift taxes | 29 |
| Federal Reserve remittances | 5 |
| Miscellaneous fees and fines | 42 |
| **Total revenues** | **5,235** |

| Outlays | $ billions |
|---|---|
| Social Security | 1,575 |
| Medicare (net) | 988 |
| Medicaid | 668 |
| Income security programs | 397 |
| Other mandatory (net) | 539 |
| Defense discretionary | 893 |
| Nondefense discretionary | 980 |
| Net interest | 970 |
| **Total outlays** | **7,010** |

**Budget balance: $5,235 − $7,010 = −$1,775 billion** (deficit)

A financing shortfall—not a measure of the effect on GDP.

Source: CBO, FY2025 Federal Budget; Revenues; 2026 Outlook. Components may not sum because of rounding. Medicare and other mandatory outlays are net of offsets.

# What Does Federal Spending Pay For?

### Public services

National defense, federal courts, and law enforcement.

### Health care and local services

Medicare, Medicaid, veterans' care, and grants for roads and education.

### Income support

Retirement and disability benefits, food assistance, and refundable tax credits.

### Interest on federal debt

Payments to lenders; these do not buy current goods or services.

Source: CBO, Federal Budget and spending breakdowns, FY2025

# Roadmap

## Policy goals and instruments

Purchases, household taxes, and transfers

## Demand and policy limits

Expansion, contraction, criticisms, and delays

## The multiplier

Initial spending and successive consumption rounds

## Stabilizers and budget balances

Automatic responses, deficits, and sustainability

# Fiscal Policy and Stabilization

### Definition

Fiscal policy uses government purchases, taxes, and transfers to influence aggregate demand.

### How it works

Purchases change G directly; household taxes and transfers change disposable income and C.

### Stabilization goal

Support output and employment in a downturn; restrain excess demand and inflation pressure.

Expansionary policy shifts AD right; contractionary policy shifts AD left.

# Fiscal Instruments

| Instrument | Example | Immediate channel |
|---|---|---|
| **Purchases (G)** | Road maintenance | A purchase of current output |
| **Transfers (TR)** | Unemployment benefits | Household disposable income |
| **Household taxes (T)** | Personal income taxes | Household disposable income |

Transfers are not G. Public infrastructure investment is G; I is private investment.

GDP accounting: BEA

# Fiscal Policy and Aggregate Expenditure

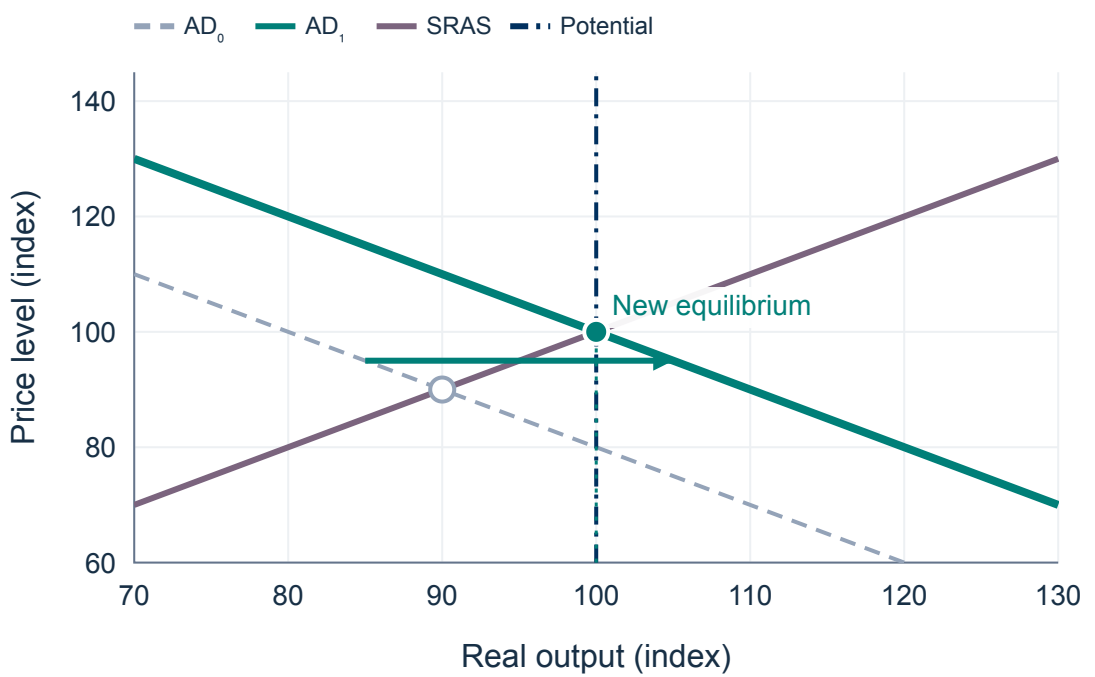
$$Y = C + I + G + NX$$

| | |
|---|---|
| **C: Consumption** | Household taxes and transfers change disposable income.<br>Tax cuts or higher transfers can raise C. |
| **I: Private investment** | Business tax incentives and expected sales affect investment.<br>Borrowing can also raise financing costs and weaken I. |
| **G: Government purchases** | Government buys current goods and services.<br>The direct expenditure channel. |
| **NX: Net exports** | Higher domestic spending can increase imports.<br>Imports reduce NX, other things equal. |

# Expansionary Fiscal Policy

## Starting output = 90 · Potential output = 100

### Expansionary · AD increases




**Recessionary gap**

Output is below potential.

**Policy change**

Government purchases increase.

**Demand response**

AD shifts right; production rises.

**New equilibrium**

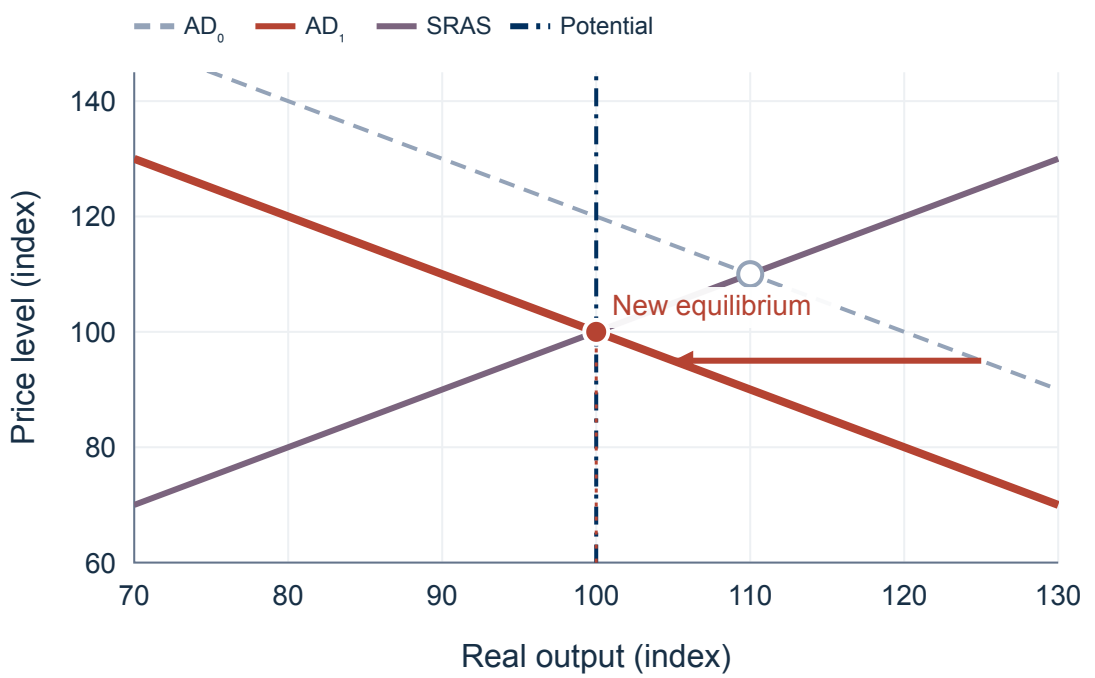
Output reaches 100; the price level rises.

## Expansionary alternatives: G ↑, household T ↓, or TR ↑.

# Contractionary Fiscal Policy

**Starting output = 110 · Potential output = 100**

## Contractionary · AD decreases




### Inflationary gap

Output is above potential.

### Policy change

Government purchases decrease.

### Demand response

AD shifts left; production falls.

### New equilibrium

Output returns to 100; the price level falls.

**Contractionary alternatives: G ↓, household T ↑, or TR ↓.**

# Fiscal Policy Checkpoint

Identify the policy direction and the immediate spending channel.

**A. Output is below potential; government repairs more roads.**

Expansionary. Purchases directly increase G.

**B. Output is above potential; household income taxes rise.**

Contractionary. Disposable income and consumption fall.

# Three Criticisms of Fiscal Expansion

| Criticism | Mechanism | Qualification |
|---|---|---|
| **Investment crowding out** | Borrowing can raise interest rates and reduce I. | The size depends on economic and monetary conditions. |
| **Consumption offset** | Current tax financing can reduce disposable income and C. | The reduction need not cancel the extra G. |
| **Ricardian equivalence** | A tax cut may be saved for expected future taxes. | A conditional tax-timing argument, holding spending fixed. |

An offset can weaken a policy without eliminating its effect.

References: CBO fiscal-policy framework; CRS fiscal-policy review

# Fiscal Policy Lags: The 2008–2009 Response

**Recognition**

Diagnose the shock.

**Decision**

Choose and authorize a response.

**Implementation**

Deliver purchases or payments.

**15 Sep 2008**

Lehman bankruptcy

**26 Jan 2009**

Recovery bill introduced

**17 Feb 2009**

ARRA signed

**May 2009**

$250 benefit payments

Authorization and delivery occur at different times.

Sources: Federal Reserve History; Congress.gov (H.R. 1); Social Security Administration

# The Spending-Round Multiplier

### Start with initial expenditure

Identify the spending on current output caused directly by the policy.

### Multiply the initial spending

With MPC = 0.8, the spending-round factor is $m = 1/(1 - MPC) = 5$.

### Compare the total multiplier effect

Net output change: $\Delta Y$ = initial expenditure change × m.

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The same m can give different output effects when initial spending differs.

# Government Purchases: Successive Spending Rounds

**ΔG = +$100 billion · MPC = 0.8 · Fixed price level**

## Government purchases

+$100 billion

Initial expenditure: ΔAE = +$100 billion.

**First spending**

$100 billion

**Next consumption round**

0.8 × $100 = $80 billion

**Further consumption rounds**

$64 billion + $51.2 billion + …

Net output effect: ΔY = $100bn × 5 = +$500bn

Spending-round factor: m = 1/(1 − 0.8) = 5

Fixed taxes and transfers; no imports or investment offsets.

# A Household Tax Cut

**Household taxes fall by $100 billion · MPC = 0.8**

### Disposable income rises

$\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

### Initial spending rises by $80 billion

With MPC = 0.8, households consume $80 billion and save $20 billion.

### Apply the same spending-round factor m = 5

Net output effect: $\Delta Y = \$80 \text{ billion} \times 5 = +\$400$ billion.

### Keep the denominator clear

m = 400/80 = 5; output per $1 of tax cut = 400/100 = 4.

# Purchases, Tax Cuts, and Transfers

**MPC = 0.8 · Change one instrument at a time**

| Expansionary change | First channel | Initial expenditure |
|---|---|---|
| **G increases by $100 billion** | Government purchases | +$100 billion in G |
| **Household T falls by $100 billion** | Disposable income +$100 billion | +$80 billion in C |
| **TR increases by $100 billion** | Disposable income +$100 billion | +$80 billion in C |

Same spending-round factor m = 5; the policy starts with $100 in G or $80 in C.

# Comparing Multiplier Effects

**MPC = 0.8 · m = 1/(1 − MPC) = 5 · Fixed lump-sum taxes**

| Policy change ($bn) | Initial ΔAE ($bn) | Common factor m | Net ΔY ($bn) | Net $\Delta S_g$ ($bn) |
|---|---|---|---|---|
| **ΔG = +100** | +100 in G | 5 | +500 | −100 |
| **ΔT = −100** | +80 in C | 5 | +400 | −100 |
| **ΔTR = +100** | +80 in C | 5 | +400 | −100 |

## Output multiplier effect: ΔY = initial ΔAE × m

Same m = 5. Output per $1 of policy: purchases 5; tax cuts and transfers 4.

Net budget change: $\Delta S_g = \Delta T - \Delta G - \Delta TR$.

A $500bn output gap: ΔG = $500bn ÷ 5 = $100bn.

One policy change at a time; fixed prices, no imports or interest/investment offsets.

# Income Taxes and the Spending Multiplier

**MPC = 0.8 · Income-tax rate t = 25% · Fixed price level**

| Additional output | | Taxes and disposable income | | Additional consumption |
|---|---|---|---|---|
| $\Delta Y = \$100$ | $\rightarrow$ | $\Delta T = \$25 \quad \Delta Y_D = \$75$ | $\rightarrow$ | $\Delta C = 0.8 \times \$75 = \$60$ |

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

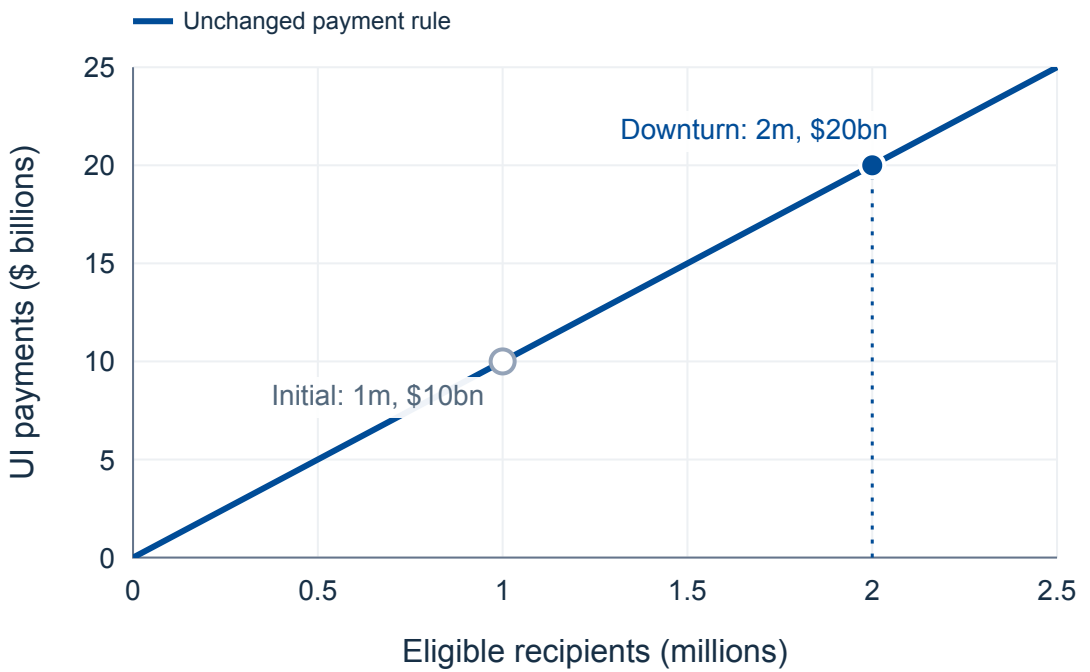
MPC remains 0.8; MPS remains 0.2.

$m = 1 / [1 - MPC(1 - t)]$

$m = 1 / [1 - 0.8(0.75)] =$ **2.5**, compared with **5** under fixed taxes.

# Automatic Stabilizers

## Unemployment insurance (UI): $10,000 per eligible recipient in this period




### Unemployment rises

More workers qualify under the same benefit rules.

### Transfers rise automatically

1m → 2m recipients; UI payments rise $10bn → $20bn.

### Consumption is cushioned

Benefits support disposable income and C; no new law is needed.

Income taxes also stabilize: falling income reduces tax payments under existing rules.

Mechanism: CBO, Automatic Stabilizers (2024); graph values are model inputs.

# Government Budget Balance

$$S_g = T - G - TR$$

T: tax revenue  G: purchases  TR: transfers

| Positive balance | Zero balance | Negative balance |
|---|---|---|
| Surplus | Balanced budget | Deficit |

T = \$500bn · G = \$400bn · TR = \$150bn

$S_g$ = \$500bn − \$400bn − \$150bn = **−\$50bn**

The balance alone does not reveal the effect on output.

Simplified budget: debt interest and other budget items omitted.

# Balanced-Budget Expansion

**All amounts in $ billions · ΔG = +100 · ΔT = +100 · MPC = 0.8 · Transfers unchanged**

| Policy change | Initial ΔAE | Policy multiplier | Total ΔY |
|---|---|---|---|
| **Purchases +100** | +100 in G | $k_G = 1/(1 - 0.8) = 5$ | 5 × 100 = **+500** |
| **Taxes +100** | −0.8 × 100 = −80 in C | $k_T = -0.8/(1 - 0.8) = -4$ | −4 × 100 = **−400** |
| **Net effect** | 100 − 80 = **+20** | Both policy effects combined | 500 − 400 = **+100** |

First round: **+$20bn**. All spending rounds: ΔY = 20 × 5 = **+$100bn**.

Budget balance: $\Delta S_g = \Delta T - \Delta G - \Delta TR = 100 - 100 - 0 =$ **$0**

**Balanced but expansionary:** extra taxes cover extra purchases, while real GDP rises.

Fixed lump-sum taxes and prices; spare capacity; no imports or interest/investment offsets.

# The Budget and the Business Cycle

**Existing tax schedules and benefit rules · No new legislation**

| Budget item | Activity weakens | Activity strengthens |
|---|---|---|
| **Tax revenue** | Usually falls | Usually rises |
| **Unemployment benefits** | More eligible claims | Fewer eligible claims |
| **Budget balance** | Moves toward deficit | Moves toward surplus |

Moving toward surplus can mean a smaller deficit.

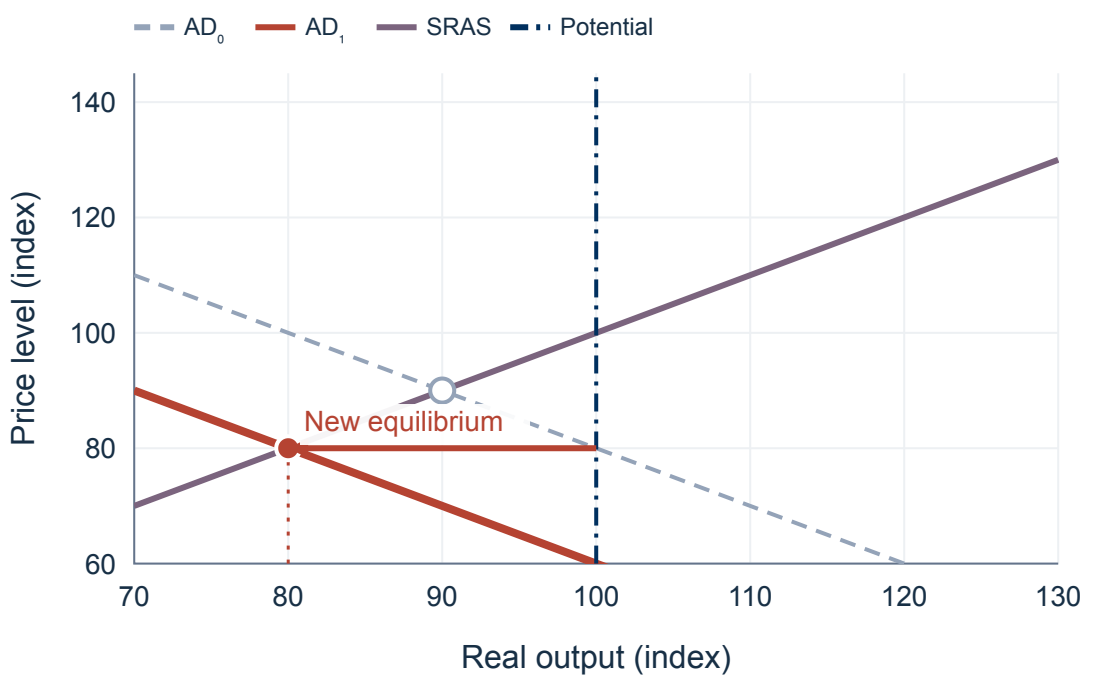
- Cyclically adjusted budget balance: estimates the balance at potential output, helping separate business-cycle effects from other changes in the budget.

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

# Annual Budget Balance and Stabilization

**Demand recession · Starting output = 90 · Potential output = 100**

**Contractionary · AD decreases**




### Initial recession

Output is already below potential.

### Annual-balance requirement

Falling receipts trigger tax increases or spending cuts.

### Demand response

The offset reduces AD further.

### Deeper output gap

Output falls from 90 to 80.

**A strict annual-balance rule can work against automatic stabilizers.**

# Fiscal Stabilization and Sustainability

### Downturns

Fiscal space allows stabilizers and temporary support to operate.

### Stronger economic activity

An opportunity to reduce deficits and rebuild fiscal space.

### Debt over time

Assess debt relative to GDP, interest costs, growth, and future balances.

A deficit is an annual flow; debt is a stock accumulated over time.

# Fiscal Policy: Final Checkpoint

Explain the mechanism and the evidence you would need.

**A. Income falls; the tax schedule stays unchanged.**

Automatic stabilization: tax payments fall.

**B. A new law raises weekly unemployment benefits.**

Discretionary expansion: transfers increase.

**C. With MPC = 0.8, compare a $100 purchase and a $100 tax cut.**

Purchases: $\Delta Y = \$500$. Tax cut: $\Delta Y = \$400$; the first round is smaller.

**D. The deficit rises in a recession. Does this prove new stimulus?**

Check new laws, automatic stabilizers, and the cyclically adjusted balance.

# Fiscal Policy Cheat Sheet: Demand

Expenditure: C + I + G + NX. Purchases enter G; household taxes and transfers influence C; business taxes can influence I.

| Policy | Main mechanism | Curve | Short-run result |
|---|---|---|---|
| **Increase purchases** | Government buys more current goods and services. | AD right | Output and price level rise. |
| **Cut household taxes** | Disposable income rises; households spend part. | AD right | Output and price level rise. |
| **Increase transfers** | Recipients spend part of the extra income. | AD right | Output and price level rise. |
| **Cut purchases or transfers; raise household taxes** | Spending falls. | AD left | Output and price level fall. |

First three: expansionary. Fourth: contractionary. Transfers themselves are not G.

Accounting: BEA · Demand mechanism: CBO

# Fiscal Policy Cheat Sheet: Supply

Fiscal policy can also affect firms' costs and the economy's productive capacity.

| Policy | Main mechanism | Possible curve shifts | Conditional result |
|---|---|---|---|
| **Productive infrastructure, education or research** | Purchases support demand now; successful investment improves capacity later. | AD right initially; possibly SRAS and LRAS right later. | Potential output can rise. |
| **Lower production taxes or subsidize inputs** | Lower marginal production costs can encourage supply. | Potentially SRAS right. | Output rises and price level falls from the supply effect alone. |

Design and financing matter. Public spending must improve productivity to raise capacity.

Fiscal framework: CBO, How CBO Analyzes Fiscal Policies

# Demand Shocks: Adjustment and Policy

Y* = potential output. Automatic adjustment below assumes no policy response and unchanged potential.

| Initial shock | Immediate result | Automatic adjustment | Fiscal response to the gap | Policy consequence |
|---|---|---|---|---|
| **AD left** | Y < Y*; price level falls. | Wages and costs fall; SRAS moves right. | Expansionary policy moves AD right. | Supports output and offsets some downward price pressure. |
| **AD right** | Y > Y*; price level rises. | Wages and costs rise; SRAS moves left. | Contractionary policy moves AD left. | Returns output toward potential and eases price pressure. |

Self-adjustment changes SRAS through wages and costs; fiscal stabilization usually changes AD.

# Supply Shocks: Adjustment and Policy

Y* = potential output. These two cost shocks leave potential unchanged; adjustment assumes no policy response.

| Initial shock | Immediate result | Automatic adjustment | Fiscal response to the gap | Policy consequence |
|---|---|---|---|---|
| **SRAS left** | Y < Y*; price level rises. | Lower wages or reversal of the cost shock can move SRAS right. | Expansionary policy moves AD right. | Supports output, but raises the price level further. |
| **SRAS right** | Y > Y*; price level falls. | Higher wages or reversal of the favorable shock can move SRAS left. | Contractionary policy moves AD left. | Returns output toward potential, but lowers the price level further. |

After an adverse supply shock, demand policy faces an output–price tradeoff.

# Potential Output and Policy Tradeoffs

### Productive capacity changes

LRAS shifts; SRAS may also shift. Adjustment is toward the new potential output.

### Choose the relevant benchmark

Fiscal choices depend on the remaining gap and the policy objective. Restoring the old GDP level may be inappropriate.

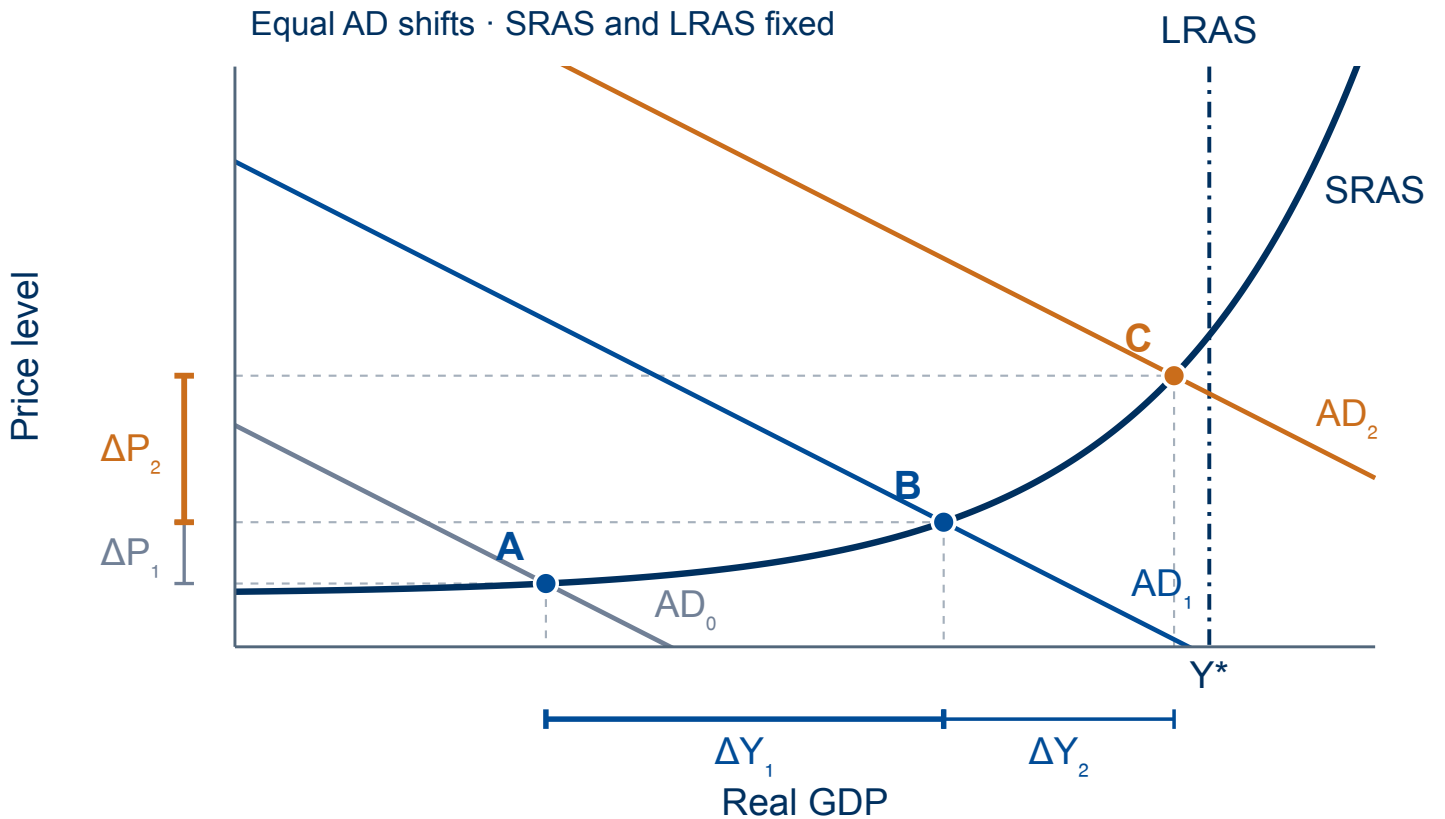
### Keep the two mechanisms distinct

Wage and cost adjustment shifts SRAS; fiscal stabilization usually shifts AD.

Results depend on timing, financing, household spending, productivity, and central-bank responses.

# The limits of continued fiscal expansion

As spare capacity disappears, additional demand generates smaller output gains and stronger inflation pressure.




### Substantial spare capacity

Firms employ idle resources. Output rises considerably, with limited price pressure.

### Capacity becomes tighter

Overtime, scarce workers, and production bottlenecks raise costs. Output responds less and prices respond more.

### Continued expansion

Persistent demand growth beyond productive capacity strengthens inflation pressure. Higher interest rates may further reduce the output response.

**Fiscal expansion becomes less effective at increasing real output as supply constraints tighten.**

Oxford College of Emory University

# Chapter 13, Part 2

## The Long-Run Implications of Fiscal Policy

ECON_OX 112 · Principles of Macroeconomics

Guilherme Paiva Pinto

# Roadmap

### Budget balances and financing

Deficits, public debt, and refinancing risk.

### History and sustainability

Greece, U.S. data, and debt relative to GDP.

### Investment and commitments

Productive capacity, interest, and future spending.

### Money and banks

Real resource limits, followed by Chapter 14.

# Deficits, Surpluses, and Public Debt

> **Deficit: a flow**
>
> Spending, including net interest, exceeds revenue during a period.

> **Surplus: a flow**
>
> Revenue exceeds spending during a period.

> **Debt: a stock**
>
> Outstanding borrowing at a date; debt held by the public excludes federal accounts' holdings.

In a simplified budget, next year's debt = this year's debt + the total deficit.

Definitions: U.S. Treasury, public debt FAQs

# Debt Service and Refinancing Risk

### Debt service

Interest payments use revenue or require additional borrowing.

### Refinancing

Maturing debt must be repaid or replaced; lenders may demand higher rates or stop lending.

### Possible feedback

Higher rates can increase borrowing and later interest costs; this is not automatic.

Crowding out affects private investment; debt service and refinancing affect the public budget.

Framework: CBO, economic effects of fiscal policy

# Greece: The Sovereign Debt Crisis

**2009: fiscal credibility**

Revisions revealed earlier misreporting and much larger fiscal imbalances.

**2010: financing pressure**

Borrowing became harder and more expensive as investor confidence fell.

**May 2010: assistance and adjustment**

Euro-area partners and the IMF supported a program tied to fiscal adjustment and reforms.

Greece used the euro; its national government did not control an independent currency.

Sources: IMF, Greece report 10/110; European Commission, assistance chronology

# The Debt-to-GDP Ratio

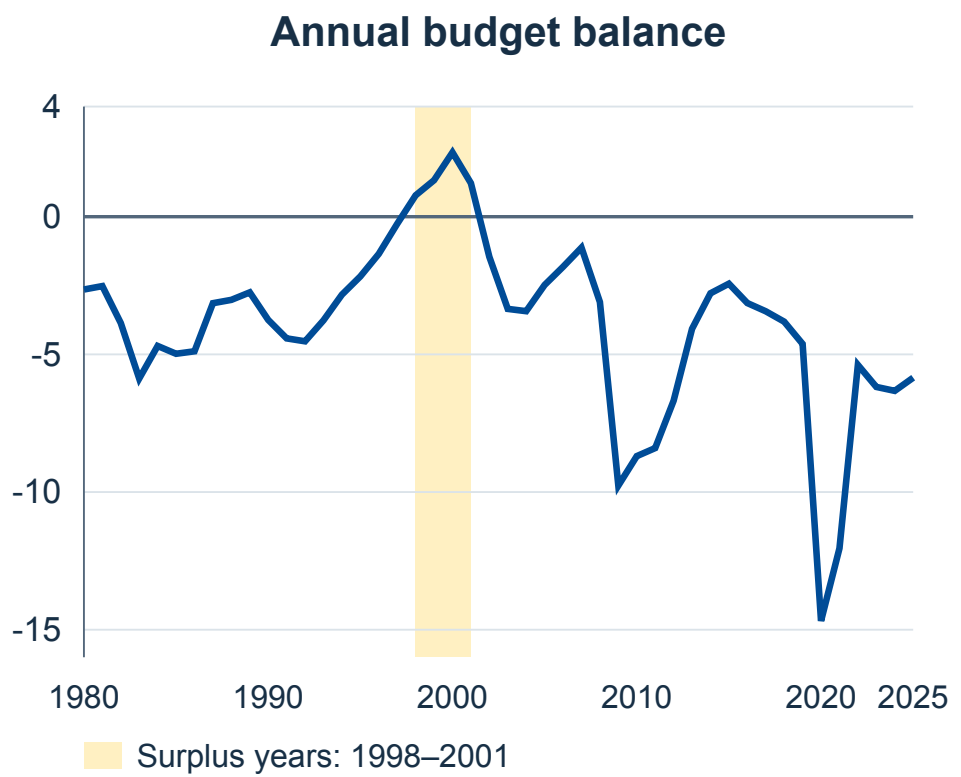
**All amounts in $ billions · Debt and GDP use current dollars**

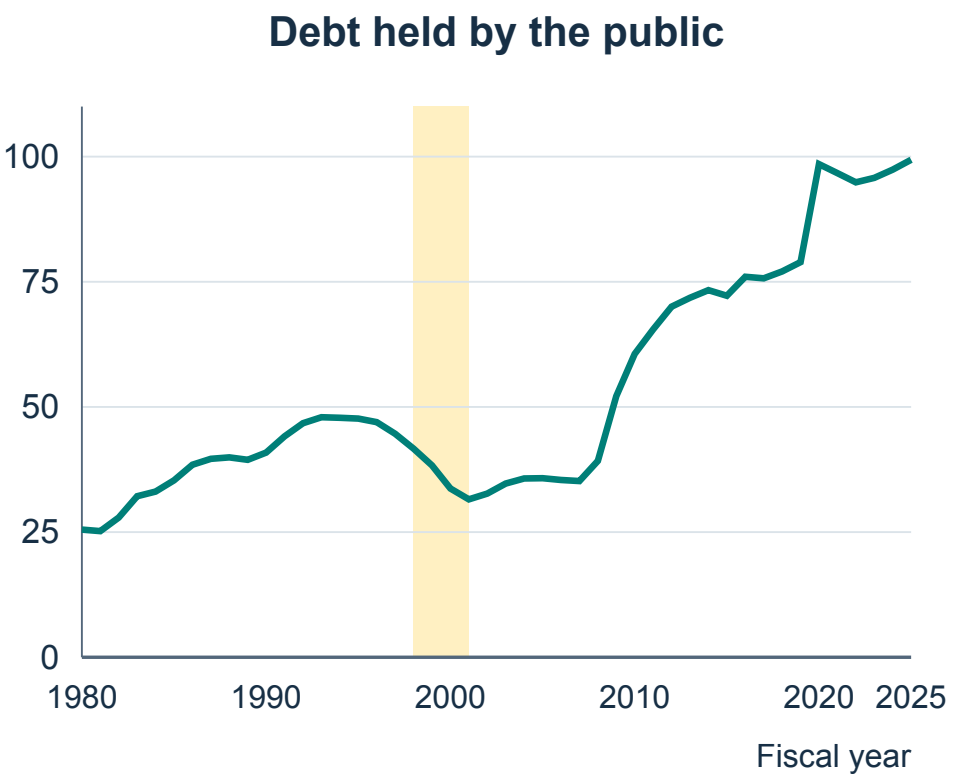
| Measure | Initial year | Following year |
|---|---|---|
| **Debt at year-end** | 600 | 620 |
| **Annual nominal GDP** | 1,000 | 1,050 |
| **Debt / GDP × 100** | 60.0% | 59.0% |

Debt rises by 20, but GDP grows faster in percentage terms, so the ratio falls.

# U.S. Budget Balances and Public Debt

**Federal government · Fiscal years 1980–2025 · Percent of GDP**




Data: CBO, Historical Budget Data, February 2026; original course chart

# Borrowing, Investment, and Growth

### Private investment

With other saving unchanged, government borrowing can raise financing costs and reduce private investment.

### Future productive capacity

Less capital accumulation can lower future productivity and potential output.

### Public investment

Effective infrastructure, education, or research can support capacity; design and financing determine the net effect.

A short-run increase in demand and a long-run increase in productive capacity are different claims.

Framework: CBO, economic effects of fiscal policy

# Interest Costs and the Primary Balance

**Amounts in $ billions · Beginning debt = 600 · Annual interest rate = 4% · Primary deficit = 10**

### Primary deficit

Noninterest spending minus revenue. The total deficit also includes net interest.

### Interest cost

0.04 × 600 = 24.

### Total deficit

24 + 10 = 34.

End-of-year debt = 600 + 34 = 634. More interest narrows other budget choices.

Budget concepts: CBO, Budget and Economic Outlook, February 2026

# Implicit Liabilities and Future Commitments

**Expected future program spending is distinct from outstanding Treasury debt.**

### Social Security

Retirement, disability, and survivor benefits; population ageing affects financing.

### Medicare

Health coverage mainly for older people; ageing and medical costs affect spending.

### Medicaid

Health coverage linked to eligibility, enrollment, costs, and policy choices.

Mandatory spending follows existing law; future costs depend on policy and assumptions.

Fiscal outlook: CBO, Budget and Economic Outlook, February 2026

# Long-Run Fiscal Policy Checkpoint

Explain the distinction before giving a policy conclusion.

**A. The annual deficit falls from 40 to 20 and remains positive. What happens to debt?**

Debt continues rising, but more slowly in the simplified budget.

**B. A deficit finances a new transport network. Does higher G prove a permanent increase in potential output?**

No. Capacity gains depend on usefulness, implementation, maintenance, and financing costs.

# Fiscal Financing and Money Creation

### A different financing method

Issuing money changes financial claims; it does not create workers, machinery, or goods.

### Real resource limits

Persistent money-financed demand beyond capacity can create inflation pressure.

### Institutional limits

Greece shared the euro; Treasury borrowing and central-bank operations are different actions.

Chapter 14: What is money, who creates it, and what role do banks play?